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Current Biology

Songbird connectome reveals tunneling of migratory neurons in the adult striatum

Highlights

- EM connectome reveals migratory neurons (MIGRs) and their interactions
- Contacts between MIGRs and mature neurons are frequent and complex
- MIGRs deform the surrounding neuropil and mature neuron somas
- MIGRs may disrupt existing circuits, a potential cost of adult neurogenesis

Authors

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In brief

Shvedov et al. use electron microscopy-based connectomics to characterize migrating immature neurons in the adult songbird striatum. These new neurons make extensive contacts with mature neurons and appear to physically deform surrounding axons, dendrites, and cell bodies, suggesting that neuron migration may physically reshape established circuitry in the adult brain.

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Songbird connectome reveals tunneling of migratory neurons in the adult striatum

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SUMMARY

Immature neurons in the adult brain migrate into existing circuits, contributing to plasticity, learning, and complex behaviors. While prior studies have examined the molecular mechanisms and functional consequences of adult neurogenesis, few have investigated the physical interactions between migrating neurons and their surrounding microenvironment. Here, we used electron microscopy (EM)-based connectomics to examine how migrating neurons interact with mature circuit elements in the adult zebra finch striatum. Migratory neurons contacted diverse structures in their microenvironment, including the axons, dendrites, synapses, and somas of mature neurons. Surprisingly, these interactions were structurally complex, often involving pronounced deformations of mature somas and the surrounding neuropil. These deformations appeared as "tunnels" made by the migratory neurons as they displaced mature structures along their path. Together, these findings suggest that migrating neurons may physically reshape the mature circuit to reach their targets, revealing an unexpected degree of structural and functional plasticity in the adult brain.

INTRODUCTION

Adult neurogenesis is believed to play an important role in learning and memory, tissue homeostasis, and regeneration.[1–3] In many adult vertebrates, this process involves migration, in which immature neurons move from proliferative zones to their integration targets. Mature brain tissue poses several challenges for migrating neurons. Compared with the developing brain, adult brains have increased cell density and diminished extracellular space,[4] and they are rich in stable synapses, which are believed to form the basis of behavior and long-term memory.[5,6] Given these constraints, how migratory neurons can physically navigate through the adult brain environment is not clear.

Songbirds are valuable model organisms for the study of neuron migration in the adult brain.[7,8] In these species, new neurons integrate into brain regions that control complex learned behaviors, where they establish synapses with mature neurons and respond to sensory stimuli.[9–13] New neurons are added to at least two subregions within the circuit that control learning and production of song: HVC (proper name), a pallial premotor nucleus, and Area X, a region within the striatum.[14]

New neurons that migrate to the striatum are born within a region of the lateral ventricle, corresponding to the mammalian medial ganglionic eminence (MGE).[15,16] After arrival in the striatum, they differentiate into medium spiny neurons (MSN) and integrate into the local circuit.[17–19] New neurons have also been reported in the adult mammalian striatum,[20–22] which has motivated further study of neurogenesis in this region in songbirds. However, a key problem is how the new neurons interact with mature circuit structures in the striatum, such as axons, dendrites, and neuronal somas. It is not known whether neurons pursue migratory routes that flexibly avoid these structural obstacles[23] or deform surrounding tissue to reach their targets.[24]

Here, we leverage the ultrastructural resolution and dense labeling of electron microscopy (EM) connectomics to investigate the microenvironment of migratory neurons in the adult brain. We focused on the first fully reconstructed zebra finch connectome, which was collected from the song nucleus Area X in the striatum.[25,26] Within this dataset, we identified a population of migratory neurons using a supervised machine-learning classifier based on morphological features derived from fluorescence microscopy. These cells had ultrastructural features consistent with migratory neurons,[27–32] such as elongated nuclei, a centriole pair at the base of the leading process, and filopodia-like extensions at the end of the leading process.

This rich dataset revealed several unexpected phenomena. Migratory neurons were observed in a synapse-dense neuropil and made extensive contacts with various features of mature neurons including spines, axons, and dendrites. Migratory neurons also made frequent soma-soma contacts with mature neurons, which form large soma clusters in the avian striatum. Interactions between new neurons and mature neurons were

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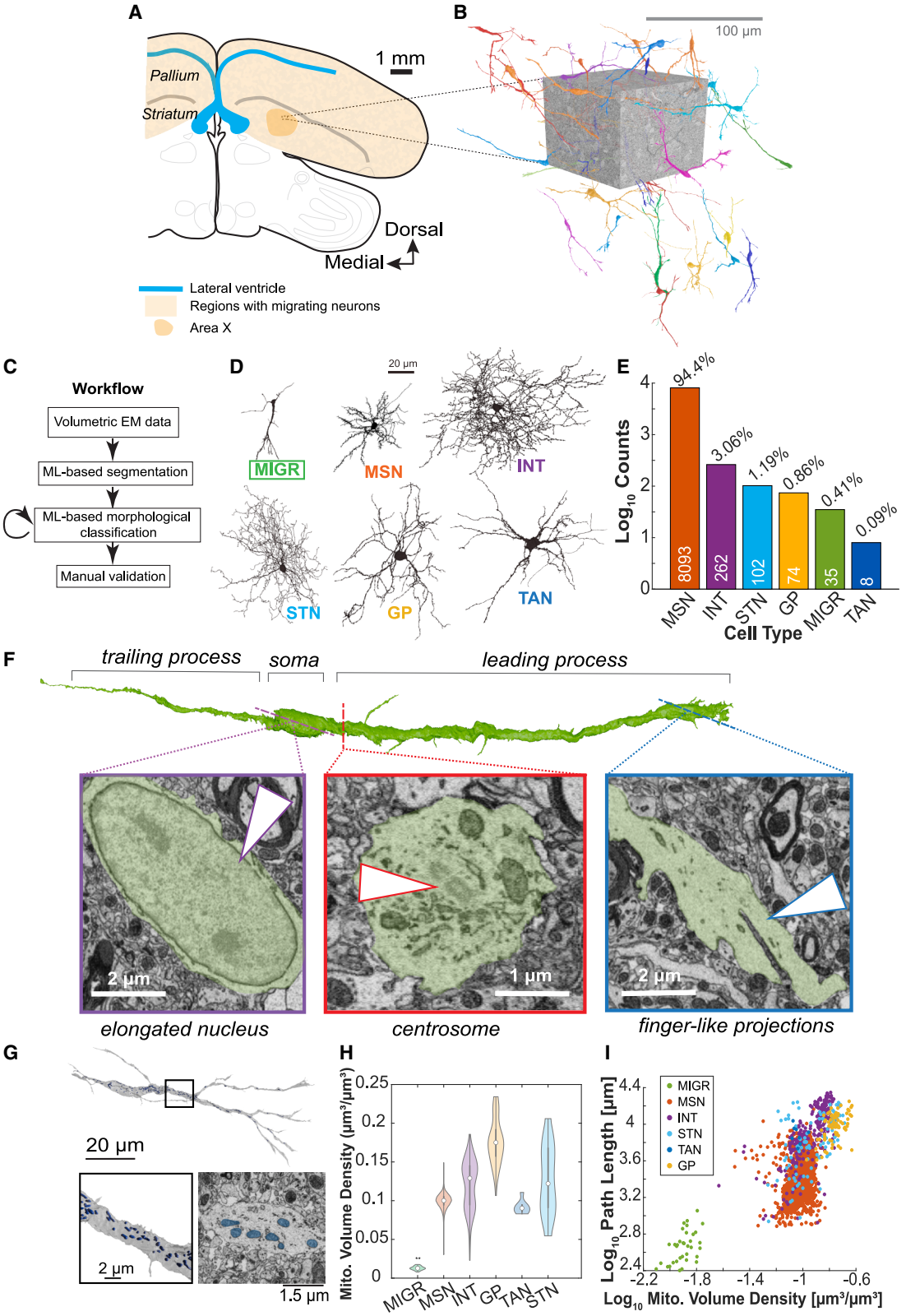


Figure 1. Identification of migratory neurons in volumetric EM

(A) Overview of neuron migration in the adult zebra finch brain. Neurons are born in the ventricular wall (blue line) and migrate throughout the pallium and striatum (orange shading), including Area X (orange outline).

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asymmetrical, with migratory neurons deforming adjacent mature neurons and neuropil. In some instances, these indentations were so extreme that the new neurons appeared to tunnel through or into clusters of mature neurons. These results reveal the value of applying EM connectomics to adult neurogenesis and suggest that migratory neurons may dramatically perturb the existing functional circuits as they migrate and integrate. Furthermore, they reveal the remarkable structural flexibility of mature neural circuits.

RESULTS

Migratory neurons identified by morphological and ultrastructural features in the EM dataset

To identify migratory neurons in the songbird basal ganglia connectome,[25] we created a library of ground-truth morphologies of new neurons in the striatum using fluorescence microscopy in tissue sections (Figure S1; see STAR Methods). We used transgenic songbirds in which new neurons are sparsely labeled with GFP,[33] allowing for the visualization of cellular morphology, combined with immunohistochemistry for double-cortin (DCX), a marker for migratory neurons,[34,35] or Hu to label neuronal identity.[27,36] These cells had two or more processes and possessed a spindle-shaped soma with a thinner "trailing" process behind the soma, and a thicker leading process in front of the soma, which was often bifurcated (Figure S1). These cells resembled migrating neurons in other regions of the adult avian forebrain[37] and in the developing mammalian brain.[38,39]

We used a machine-learning-based classifier, trained on a subset of manually identified cells, to identify other migratory neurons from the dataset (Figures 1A–1D and S1; see STAR Methods). This approach defined a cell class consisting of 135 objects. Manual inspection revealed that 40 objects within this class met criteria for migrating neurons (false positives = 95/135; false negatives = 0; see STAR Methods). We selected 35 true positives with largely complete reconstructions for further analysis (Figure S2).

Identified cells had distinct morphological and ultrastructural features consistent with a migratory neuron identity (Figure 1F). All cells had an elongated soma with thin cytoplasm surrounding a nucleus containing one or more nucleoli with heterochromatin-dense edges, as reported previously for migratory neurons across species and developmental stages.[27–32] In 34/35 of these cells, the centrosome was clearly located below the nucleus at the base of the leading process, as expected for migratory neurons.[28,40,41] A Golgi apparatus was typically found near the centrosome, which has been reported to be aligned with the direction of neuron migration in immature neurons.[42–44] All had one or more "finger-like" projections at the distal tips of their leading processes, similar to the filopodia or growth-cone structures described for migrating neurons and extending neurites.[45] Finally, these cells typically had small, rounded mitochondria, which were clustered around the centrosome, as previously described for immature and migratory neurons[46,47] (Figure 1G). They also exhibited a significantly lower mitochondrial density than that of mature neurons (Figures 1G–1I), which was perhaps indicative of a lower metabolic demand.[46,48–52]

Together, our findings demonstrate that the songbird basal ganglia connectome[25] contains a distinct class of cells with morphological and ultrastructural features that resemble immature neurons in the midst of or in the end stages of migration.

Migratory neurons are present at a high density and oriented in multiple directions

Next, we evaluated the orientations and positions of identified migratory neurons throughout the connectome. We defined the orientation of the cell as a 3-dimensional vector beginning at the soma and ending at the first branch point of the leading process (Figures 2A and 2B; STAR Methods). Leading process vectors from different cells were oriented in a variety of directions (Figures 2A and 2D). No directional bias was detected in the horizontal plane (Rayleigh's test, $p = 0.05$), and a slight bias was detected in the azimuth plane (Rayleigh's test, $p < 0.001$). The correspondence of the sample coordinate framework (x, y, and z) and tissue axes (dorsal, anterior, and lateral) was not recorded in the original dataset, however, so it is not possible to attribute this slight directional bias to an anatomical direction.

Identified migratory neurons were present at a density of 1,390 neurons per mm^3 (35 cells in 0.025 mm^3), and their distribution in the volume was well fit by a maximum entropy model, indicating uniform distribution throughout the tissue (Figure 2E). Our findings of high soma position entropy and multidirectional leading process orientation are similar to observations from *in vivo* imaging studies of migration[33,37] and consistent with a diffusion-like process.

Migratory cells are located within synapse-dense neuropil

Since migratory neurons were widely distributed throughout the tissue volume, we investigated whether MIGRs interacted with synapses or avoided synapse-rich areas (Figure 3A). This connectome contains over 8 million high-confidence ($\geq$0.9 certainty) synapses at a density of 0.29 per μm^3 (Figures 3B–3D). The local synapse density surrounding migratory neurons was not significantly different from the global synapse density (0.31 ± 0.07 per μm^3, one-sample t test, and $p = 0.12$) or the density of synapses surrounding random points throughout the volume (0.28 ± 0.11 per um^3, Kolmogorov-Smirnov test, and

(B) Reconstructions of migratory neurons from volumetric EM.
(C) Workflow for automated identification of migratory neurons.
(D) Tracings of representative cells from identified cell classes.
(E) Prevalence for each cell class in the dataset.
(F) Ultrastructural features (arrows) characteristic of migratory neurons from an example cell.
(G) (Top) Reconstruction of a migratory cell with segmented mitochondria in blue. (Bottom Left) Zoom in on the box region from the top image. (Bottom right) Zoom-in cross section with mitochondria in blue.
(H) Mitochondrial densities across cell types. The MIGR cell type has significantly lower mitochondrial densities than all other cell types (one-way ANOVA, $p = 0$).
(I) Path lengths and mitochondrial density volume for immature and mature neuron classes.
See also Figure S2; Table S1.

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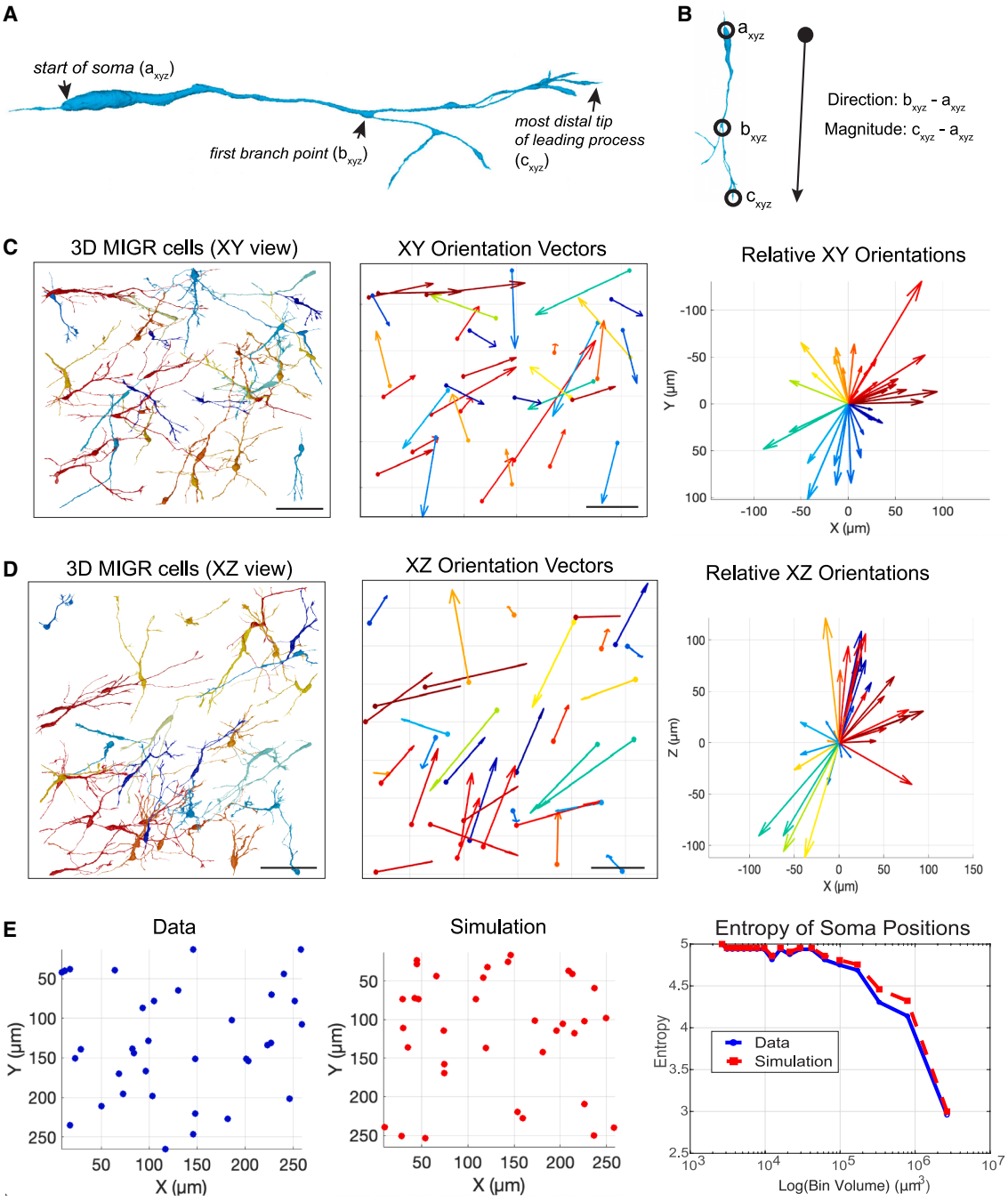


Figure 2. Migratory neurons are widely distributed and oriented in multiple directions

(A) Morphological features used for computing cell orientation.

(B) Formula for computing cell orientation.

(C) (Left) *xy* view of migratory neurons (MIGR; $n = 35$) in the volume. (Middle) *xy* orientation vectors; magnitudes of vectors vary according to the length of MIGR. Scale bars, 50 µm. (Right) Polar plot of relative *xy* vector directions with start position and magnitude normalized. No directional bias was detected (Rayleigh's test, $p = 0.05$).

(D) (Left) *xz* view of MIGR cells ($n = 35$). (Middle) MIGR *xz* orientation vectors. (Right) Polar plot of relative *xz* vector directions. Significant directional bias was detected (Rayleigh's test, $p < 0.001$).

(E) (Left) Recorded soma positions. (Middle) Simulated soma positions (maximum entropy model). (Right) Computed entropy of data (blue) vs. simulation (red) across different bin sizes.

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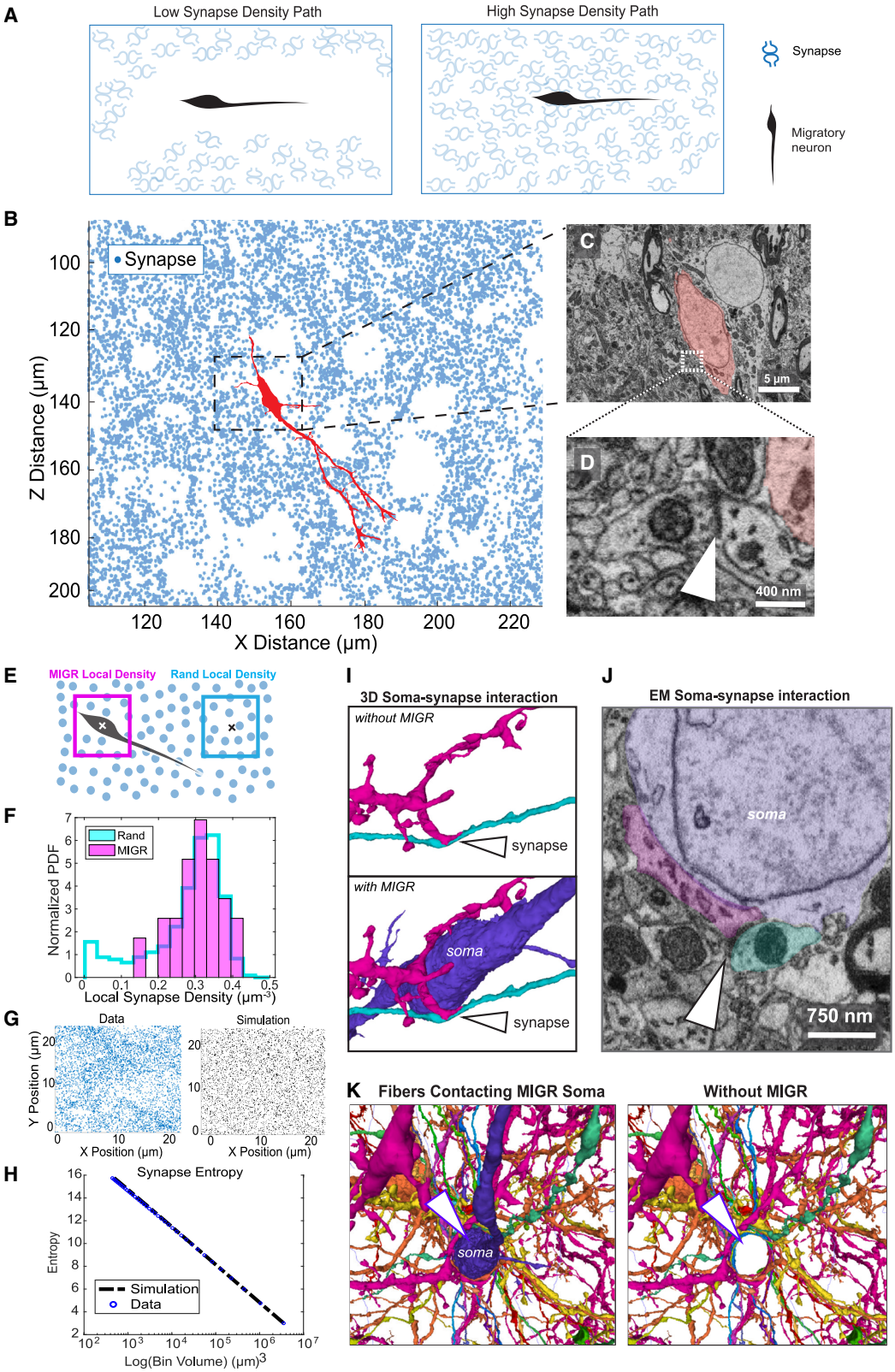


Figure 3. Migratory cells located within a synapse-dense neuropil

(A) Conceptual models; neurons may migrate through synapse-free corridors (left) or dense neuropil (right).

(B) Synapse locations (blue) surrounding a MIGR (red).

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p = 0.35; Figures 3E and 3F; STAR Methods), suggesting a roughly uniform probability distribution of synapses throughout the tissue. Indeed, the distribution of synapses throughout the entire volume was well fit by a maximum entropy model (uniform distribution, n = 8,278,615 simulated synapses) (Figures 3G and 3H).

Visual inspection of ultrastructural features revealed that migratory neurons contacted synapses between other neuron pairs (Figures 3I and 3J). Moreover, we observed dendrites of mature neurons curving around the soma of migratory cells, raising the possibility that migratory neuron somata deform neuropil in their vicinity.

To characterize the pattern of synapses around MIGRs in more detail, we measured the density at different distances from the soma (Figures S3A and S3B). As expected, MIGR somata formed cavities in the synapse density (Figures S3B and S3C). Interestingly, there was a significant increase in synapse density immediately surrounding MIGR somata compared with 15–20 μm away (Figure S3D; p = 0.0032; paired Wilcoxon signed-rank test). This perisomatic peak in synapse density was not explained by synapses onto MIGRs themselves (STAR Methods) and was not observed in mature neuron types (Figure S3E). Finally, a high density of synapses was also observed around the putative leading processes of MIGRs, and many of these synapses lay within 2 μm of the leading process, a radius smaller than the smallest observed MIGR soma. Together, these results indicate that MIGRs are surrounded by fields of dense synapses and raise the possibility that these new neurons displace existing axons, dendrites, and synapses, compressing tissue as they move.

Interactions with cellular scaffolds and somas

We next examined the interactions between new neurons and other classes of cells. In other systems, including other areas of the avian brain, many new neurons use scaffolds, such as radial fibers or blood vessels, to migrate.[29,53] However, we observed only a small minority of migratory neurons (n = 8/35) associated with known migratory scaffolds (Figure S4). In contrast, half of the identified migratory neurons formed close soma-soma associations with mature neurons (n = 17/35 migratory neurons; Figure S4) and, to a lesser extent, astrocytes (n = 4/35; Figure S5). Interestingly, there was no significant difference between the proximity of migratory neurons to mature neurons (7.91 ± 2.55 μm) and the proximity of random points throughout the volume to mature neurons (9.14 ± 4.00 μm; KS test, p = 0.13; Figures S4I and S4J). Thus, these encounters may be due to the high density of mature neurons along the paths of migrating neurons, rather than an active process of attraction or avoidance.

Migratory neurons deform mature neuron neighbors

Given the prevalence of mature-migratory neuron soma contacts, we examined their interactions in more detail. Specifically, we wondered whether migrating neurons conform their shape to the boundaries of mature neurons (Figure 4A). Examination revealed that, rather than conforming to mature neurons, migrating neurons deformed adjacent neuronal somas (n = 17 interactions; Figure 4B).

To quantify this deformation, we measured indentations at the site of contact between migrating and mature neurons (Figure 4C). These indents were large and asymmetrical between contact partners (Figure 4D). Indents were significantly deeper in mature neuron somas (2.03 ± 1.58 μm deep, 7.15 ± 5.66 μm long) than in migratory neuron somas (0.59 ± 0.30 μm deep, 1.64 ± 1.08 μm long; paired t test, $p < 0.004$; Figure 4E; STAR Methods). In several cases (n = 12), migratory neurons were found within large concavities involving one (n = 4/12) or more mature neurons (n = 8/12).

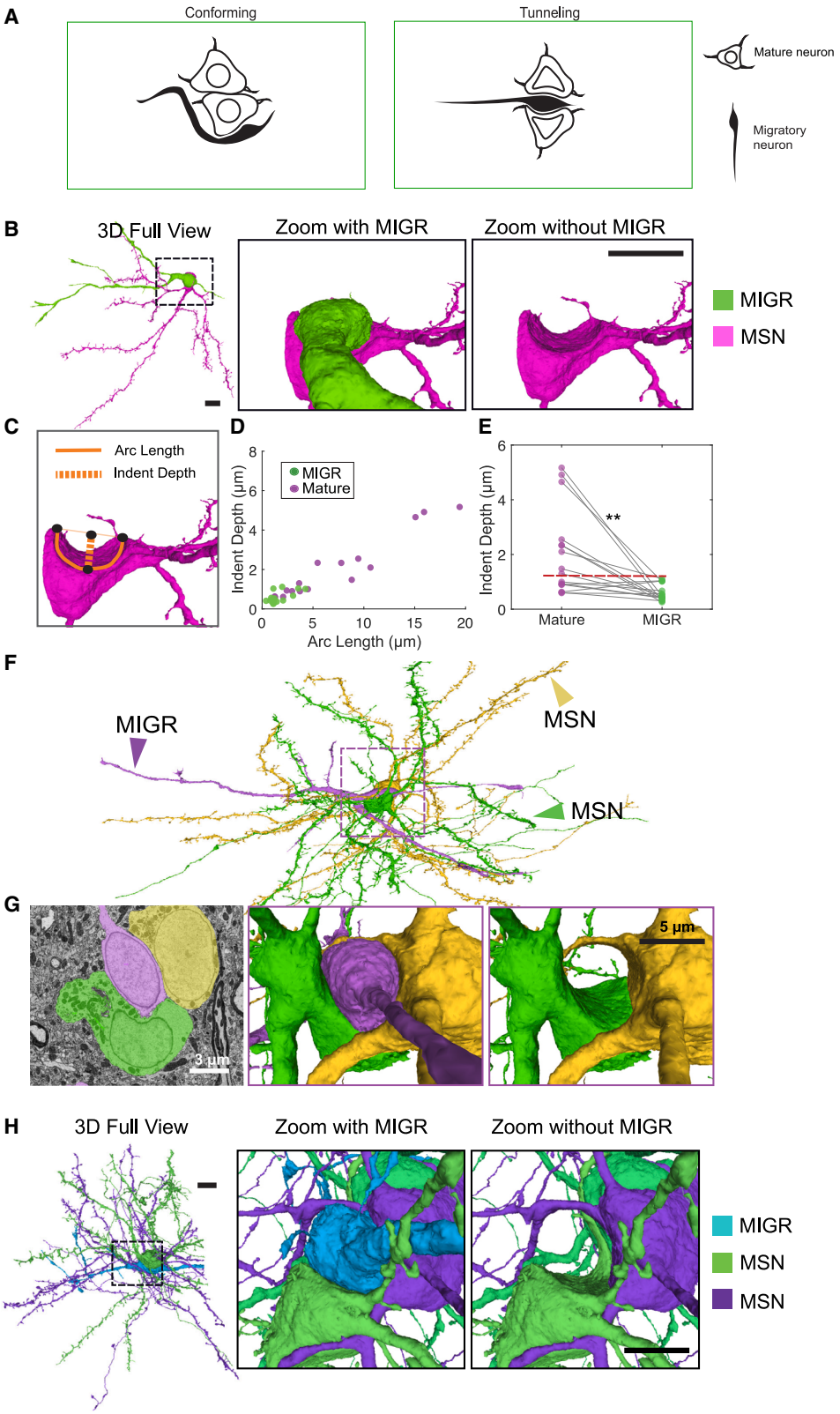
This phenomenon, in which new neurons appear to deform multiple adjacent neuron somas, axons, and dendrites, we refer to as "tunneling." Tunneling was particularly evident in migratory neuron interactions with groups of densely packed mature neurons (Figures 4F–4H and S6).

New neurons interact with clusters in the avian striatum

In other areas of the songbird forebrain, such as the HVC, mature neurons are organized into "clusters," a motif in which somas are packed together and surrounded by neuropil.[8,54–57] Whether clusters exist in the striatum and how immature neurons interact with these structures remains unknown. In the striatum, mature neurons were closer together than by chance (mean = 7.92 ± 2.78 μm; KS test, $p < 0.00001$) and frequently contacted the somas of other neurons (Figure 5). We found that 71.9% of all mature neurons (n = 6,163) were found in clusters, defined as a connected graph of adjacent neuron somas (minimum distance < 9 μm between soma centers, see STAR Methods). Clusters (n = 1,574) ranged in size from 2 to 33 neurons (mean = 4 ± 3) and contained all classes of mature neurons (Figures 5E–5H; STAR Methods) as well as glia (Figure S5). The predominant cell type in the clusters was MSNs (n = 5,786 MSNs; 93.9% of all clustered cells; Figure 5I); however, after normalization by cell frequency, all neuron types had a similar contribution to cluster composition (Figure 5J). New neurons

(C) Electron micrograph of the dashed box in (B).
(D) Zoom in on the dashed box region in (C), showing a synapse (arrow) near MIGR soma (red).
(E) Schematic for local density computation method.
(F) Histogram of local densities around MIGRs (n = 35) and random points (n = 29,484). No significant difference was found between local MIGR density and local Rand density (Kolmogorov-Smirnov test, p = 0.35).
(G) (Left) Synapse coordinate data in 25 × 25 × 10 μm field-of-view (confidence ≥ 0.9; n = 4,929 synapses). (Right) Simulated synapse coordinates in 25 × 25 × 10 μm field-of-view (n = 4,929 simulated synapses). The simulation assumed a maximum entropy model and followed a uniform probability distribution.
(H) The Shannon entropy of the synapse coordinates vs. simulated uniform distribution across varying 3D bin sizes.
(I) (Top) 3D render of dendrite processes without a MIGR cell. (Bottom) 3D render of dendrite processes with a MIGR cell.
(J) EM micrograph of a MIGR soma (from I) contacting synapse between two nearby dendrites.
(K) (Left) 3D render of a MIGR soma contacting several passing fibers. (Right) Same as (left), but without MIGR visualization, showing concavity left in surrounding fibers.
See also Figure S3.

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were found in clusters, where they indented one or more mature neurons, forming tunnels through the clusters (Figures 5K–5M). In contrast, astrocytes and other glia did not indent mature neurons but were instead indented by neurons (Figure S7).

DISCUSSION

These data reveal the intricate interactions between migratory neurons in the adult striatum and their local environment. Our findings support a model in which migrating neurons disperse throughout dense neural tissue in multiple directions, making various contacts with surrounding structures. In addition, our data reveal a previously undescribed form of neuron migration in which new neurons cause deformities in nearby neurons and synapses. Together, these data highlight the value of applying connectomics to problems in developmental neuroscience.[58,59] Future studies using EM connectomics could illuminate the later stages of new neuron maturation, capturing how synaptic contacts evolve as new neurons integrate into functional adult circuits.

Several features of the migration we observed resemble those reported in the embryonic and adult nervous system, including in humans.[30,60–64] For example, MIGRs resemble other migratory cells, such as immature interneurons in the cerebral cortex[30,31,63] and immature projection neurons in HVC,[37] both of which have a complex multipolar morphology and change direction frequently as they move. However, to our knowledge, tunneling migration by neurons has not previously been reported in the vertebrate nervous system. This may be due to methodological constraints, as this type of comprehensive high-resolution survey, provided by EM connectomics, is rarely applied to studying neuron migration in development or adulthood. Alternatively, it is possible that tunneling is a specialization of neurogenesis in this avian system. This behavior contrasts with chain migration in the rostral migratory stream,[65] where neurons move through specialized corridors segregated from mature neurons, and with the amoeboid migratory strategy observed in some glial cells, which typically conform to the contours of mature neuronal somas.[23] Interestingly, tunneling-like behavior has been described in metastatic cancer cells, which navigate confined spaces by actively deforming their microenvironments.[66] Tunneling may therefore reflect a conserved strategy adopted by specialized migratory cell types in dense tissues. Future research could uncover the molecular and biophysical mechanisms that may enable this form of migration and apply EM connectomics to evaluate its prevalence across systems and species.

Importantly, tunneling reveals a potential cost of adult neuron migration: disruption of existing circuits. As new neurons move through the mature tissues, mechanical disruption of cell bodies, axons, and dendrites could affect synaptic connections, either through breakage or acute mechanical perturbation.[67–69] Such perturbations could alter circuit-level dynamics and may necessitate compensatory homeostatic mechanisms, similar to those proposed to support injury resilience in the songbird brain.[70,71] Alternatively, such network perturbations could also provide a substrate for plasticity. Indeed, recent advances in deep reinforcement learning indicate that sparse random perturbations in network connections can be used to facilitate continual learning in artificial neural systems.[72,73]

Several limitations should be acknowledged. First, chemical fixation and EM processing can alter tissue dimensions and reduce extracellular space,[74] potentially constricting the neuropil around MIGRs; although many morphological features we report are unlikely to be explained solely by processing shrinkage, we cannot rule out some contribution of fixation-related deformation to the positional information. Second, the voxel resolution of this dataset limits the visualization of fine ultrastructural elements (e.g., gap junctions), which limits our ability to completely characterize cell-cell interactions.[25] Finally, annotation and segmentation errors and volume clipping at the edges of the imaged block can influence morphological interpretation. To mitigate this, each MIGR was inspected and validated by hand (see STAR Methods). Taken together, these factors warrant some caution in interpreting the absolute prevalence and fine-scale ultrastructure of MIGRs.

Furthermore, tunneling migration should be considered in the context of the full migratory trajectory of neurons. Neurons are born in the lateral ventricle, yet our dataset samples a restricted region within Area X, several millimeters away. This may bias the types of migration we observe. For example, long-range migration along radial glial fibers, described elsewhere in the avian brain,[37,53] could be underrepresented. Previous work has suggested multiple forms of migration, including one fast radial glial-dependent form, and the other slow and radial glial-independent.[37,53,75] Tunneling may represent a specific feature of this second form of migration.

While the observed features are consistent with migratory identity, tunneling may reflect a terminal stage of migration and

Figure 4. Migratory neurons form indentations in mature neuron somas

(A) Alternative strategies for interactions with mature somas.

(B) (Left) Example 3D render of soma-soma contact between a migratory neuron (MIGR; green) and one MSN (pink). (Middle) Zoomed-in perspective of MIGR and MSN, and (right) same zoomed-in perspective without the MIGR cell. Depth of indent is 2.15 μm. Scale bar, 5 μm.

(C) Cell indent quantification method.

(D) Indent depth and length for each associated MIGR (green) and mature neuron (purple) ($n = 17$).

(E) Indent depth on each cell for Mature-MIGR soma pairs ($n = 17$). Indent depth is significantly higher on mature neurons of each pair than their corresponding MIGR soma (paired t test, $p < 0.004$). The red line indicates median mature neuron indent depth.

(F) Full 3D render of MIGR cell (purple) in soma-soma contact with two MSNs (yellow and green).

(G) (Left) Electron micrograph of interaction from (F). (Middle) 3D render of MIGR cell (purple) and MSNs (green and yellow) from (F). (Right) 3D render of MSNs with MIGR cell removed.

(H) (Left) "Tunneling" example with full-view 3D rendering of MIGR (blue) and mature neurons (MSN class; green and purple), (middle) zoomed-in perspective of MIGR and MSNs, and (right) same perspective as in middle without MIGR. Depth of indent is 4.

See also Figures S4, S6, and S7.

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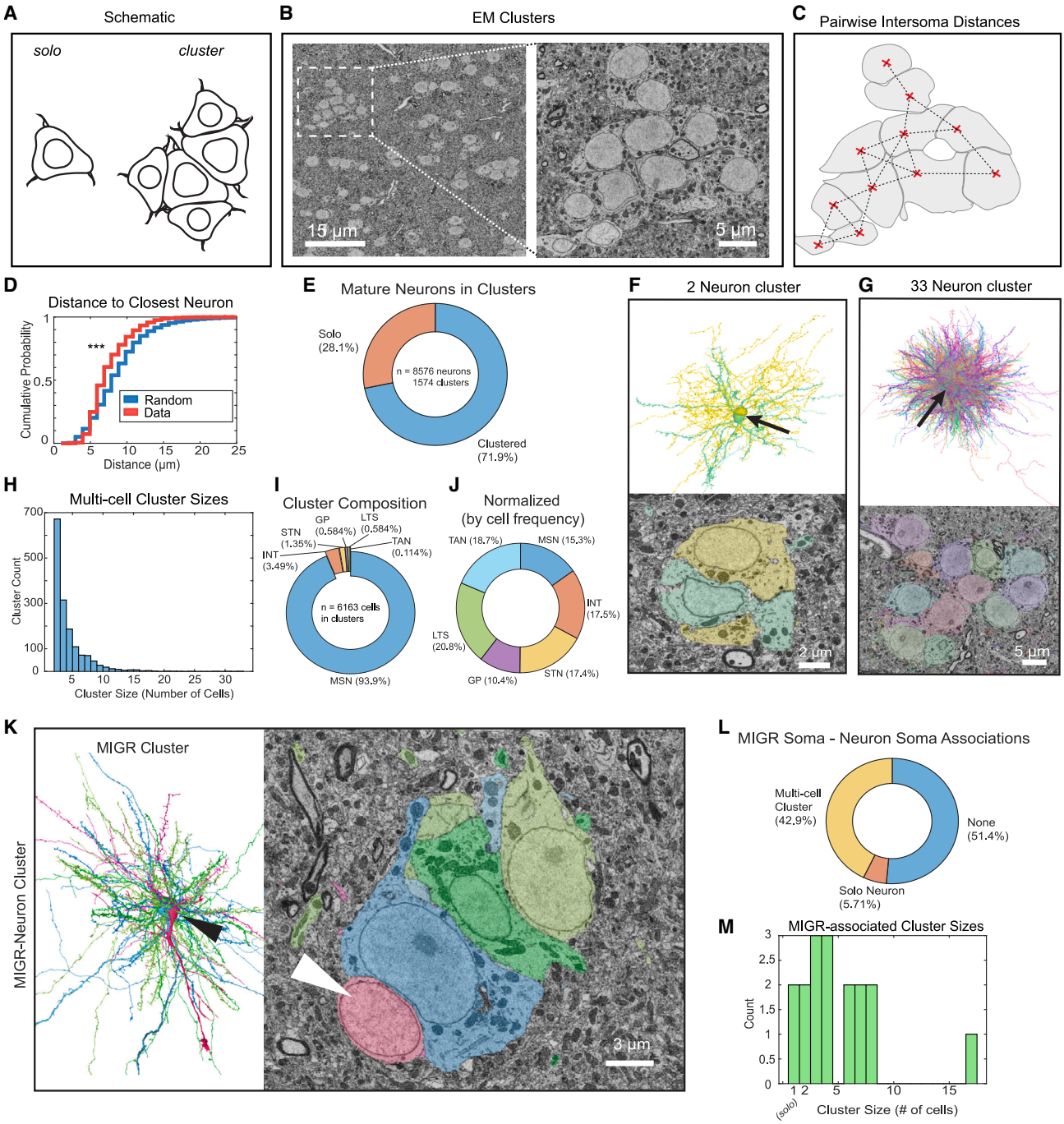


Figure 5. Migratory neurons associate with neuronal clusters

(A) Schematic of a solo neuron vs. a cluster of neurons.

(B) Single plane EM micrograph from Area X. (Right) Zoomed-in view of the dashed region.

(C) Schematic for intersomal distance quantification. Clusters were identified using nearest neighbor network analysis based on intersomal distance.

(D) Mature neurons (blue line) are closer to neuron neighbors than expected by chance (red line; Kolmogorov-Smirnov test, $p < 0.00001$).

(E) Of all mature neurons, 28.1% are on their own, and 71.9% are in clusters.

(F) Example two-neuron cluster. 3D render (top, arrow) and EM micrograph (bottom).

(G) Example 33-neuron cluster, 3D render (top, arrow) and EM plane through somas (bottom).

(H) Distribution of computed cluster sizes.

(I) Cell-type composition of all clusters.

(J) Cell-type composition of all clusters, normalized by the frequency of each cell type in the data.

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may lead to the formation of neuron clusters. Our analysis suggests that the striatum shares a nuclei-cluster organization with other regions of the songbird brain. This motif, in which brain nuclei[76,77] contain clusters of tightly packed neuronal somata,[54,78] contrasts sharply with the laminar nature of the mammalian cortex. Moreover, data hint at a bidirectional interaction between the cluster motif and adult neurogenesis.[79] In other areas of the songbird brain, new neurons preferentially stop migrating and integrate adjacent to mature ones,[8,37,54,80] a process that may drive the formation and growth of clusters. In addition, clusters may provide a permissive microenvironment, including trophic support and patterned activity, possibly via gap junctions, that promotes survival and maturation of incoming neurons.[56,57,78,80,81] Perhaps the nuclei-cluster organization of the songbird forebrain may accommodate lifelong neuronal addition in ways the laminar organization of the mammalian cortex may not.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Dr. Benjamin Scott (bbs@bu.edu).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- The raw data used in this paper are available at SyConn Web (syconn.esc.mpcdf.mpg.de). All cells analyzed and used in figures can be visualized with SyConn Web using the Cell IDs given in Table S1.
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

J.K. performed data collection, data curation, and managed data sharing. B.B.S. and J.K. conceived the project. N.R.S., A.R., S.J.C., and D.E.S. performed analysis and contributed to the figures. N.R.S. wrote the manuscript with feedback from B.B.S. All authors discussed and edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Experimental model details
- METHOD DETAILS
  - Immunohistochemistry
- QUANTIFICATION AND STATISTICAL ANALYSIS
  - Machine learning based identification of migratory neurons
  - Cell orientation analysis
  - Mitochondrial density estimates
  - Synapse density estimates
  - Shannon Entropy estimation
  - Cell somas and cluster analysis

SUPPLEMENTAL INFORMATION

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REFERENCES

1. Kempermann, G., Wiskott, L., and Gage, F.H. (2004). Functional significance of adult neurogenesis. Curr. Opin. Neurobiol. *14*, 186–191. https://doi.org/10.1016/j.conb.2004.03.001.
2. Ming, G.L., and Song, H. (2011). Adult Neurogenesis in the Mammalian Brain: Significant Answers and Significant Questions. Neuron *70*, 687–702. https://doi.org/10.1016/j.neuron.2011.05.001.
3. Chapouton, P., Jagasia, R., and Bally-Cuif, L. (2007). Adult neurogenesis in non-mammalian vertebrates. BioEssays *29*, 745–757. https://doi.org/10.1002/bies.20615.
4. Lehmenkühler, A., Syková, E., Svoboda, J., Zilles, K., and Nicholson, C. (1993). Extracellular space parameters in the rat neocortex and subcortical white matter during postnatal development determined by diffusion analysis. Neuroscience *55*, 339–351. https://doi.org/10.1016/0306-4522(93)90503-8.
5. Holtmaat, A., Wilbrecht, L., Knott, G.W., Welker, E., and Svoboda, K. (2006). Experience-dependent and cell-type-specific spine growth in the neocortex. Nature *441*, 979–983. https://doi.org/10.1038/nature04783.
6. Roberts, T.F., Tschida, K.A., Klein, M.E., and Mooney, R. (2010). Rapid spine stabilization and synaptic enhancement at the onset of behavioural learning. Nature *463*, 948–952. https://doi.org/10.1038/nature08759.
7. Alvarez-Buylla, A., and Nottebohm, F. (1988). Migration of young neurons in adult avian brain. Nature *335*, 353–354. https://doi.org/10.1038/335353a0.
8. Kirn, J.R., Fishman, Y., Sasportas, K., Alvarez-Buylla, A., and Nottebohm, F. (1999). Fate of new neurons in adult canary high vocal center during the first 30 days after their formation. J. Comp. Neurol. *411*, 487–494. https://doi.org/10.1002/(SICI)1096-9861(19990830)411:3<487::AID-CNE10>3.0.CO;2-M.
9. Goldman, S.A., and Nottebohm, F. (1983). Neuronal production, migration, and differentiation in a vocal control nucleus of the adult female

(K) Example migratory neuron (MIGR) associating with cluster. 3D renders (left) and EM micrograph (right). The arrowheads point to MIGR soma.
(L) Frequency of associations of MIGR cells with mature neuron somas.
(M) Sizes of clusters contacting migratory neurons.
See also Figure S5.

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canary brain. Proc. Natl. Acad. Sci. USA *80*, 2390–2394. https://doi.org/10.1073/pnas.80.8.2390.

10. Paton, J.A., and Nottebohm, F.N. (1984). Neurons generated in the adult brain are recruited into functional circuits. Science *225*, 1046–1048. https://doi.org/10.1126/science.6474166.
11. Walton, C., Pariser, E., and Nottebohm, F. (2012). The Zebra Finch Paradox: Song Is Little Changed, But Number of Neurons Doubles. J. Neurosci. *32*, 761–774. https://doi.org/10.1523/JNEUROSCI.3434-11.2012.
12. Alvarez-Buylla, A., Ling, C.Y., and Yu, W.S. (1994). Contribution of neurons born during embryonic, juvenile, and adult life to the brain of adult canaries: Regional specificity and delayed birth of neurons in the song-control nuclei. J. Comp. Neurol. *347*, 233–248. https://doi.org/10.1002/cne.903470207.
13. Polomova, J., Lukacova, K., Bilcik, B., and Kubikova, L. (2019). Is neurogenesis in two songbird species related to their song sequence variability? Proc. Biol. Sci. *286*, 20182872. https://doi.org/10.1098/rspb.2018.2872.
14. DeWulf, V., and Bottjer, S.W. (2005). Neurogenesis within the juvenile zebra finch telencephalic ventricular zone: A map of proliferative activity. J. Comp. Neurol. *481*, 70–83. https://doi.org/10.1002/cne.20352.
15. Scott, B.B., and Lois, C. (2007). Developmental origin and identity of song system neurons born during vocal learning in songbirds. J. Comp. Neurol. *502*, 202–214. https://doi.org/10.1002/cne.21296.
16. Puelles, L., Kuwana, E., Puelles, E., Bulfone, A., Shimamura, K., Keleher, J., Smiga, S., and Rubenstein, J.L. (2000). Pallial and subpallial derivatives in the embryonic chick and mouse telencephalon, traced by the expression of the genes Dlx-2, Emx-1, Nkx-2.1, Pax-6, and Tbr-1. J. Comp. Neurol. *424*, 409–438. https://doi.org/10.1002/1096-9861(20000828)424:3<409::AID-CNE3>3.0.CO;2-7.
17. Rochefort, C., He, X., Scotto-Lomassese, S., and Scharff, C. (2007). Recruitment of FoxP2-expressing neurons to area X varies during song development. Dev. Neurobiol. *67*, 809–817. https://doi.org/10.1002/dneu.20393.
18. Kosubek-Langer, J., Schulze, L., and Scharff, C. (2017). Maturation, Behavioral Activation, and Connectivity of Adult-Born Medium Spiny Neurons in a Striatal Song Nucleus. Front. Neurosci. *11*, 323. https://doi.org/10.3389/fnins.2017.00323.
19. Kosubek-Langer, J., and Scharff, C. (2020). Dynamic FoxP2 levels in male zebra finches are linked to morphology of adult-born Area X medium spiny neurons. Sci. Rep. *10*, 4787. https://doi.org/10.1038/s41598-020-61740-6.
20. Arvidsson, A., Collin, T., Kirik, D., Kokaia, Z., and Lindvall, O. (2002). Neuronal replacement from endogenous precursors in the adult brain after stroke. Nat. Med. *8*, 963–970. https://doi.org/10.1038/nm747.
21. Yamashita, T., Ninomiya, M., Hernández Acosta, P., García-Verdugo, J.M., Sunabori, T., Sakaguchi, M., Adachi, K., Kojima, T., Hirota, Y., Kawase, T., et al. (2006). Subventricular Zone-Derived Neuroblasts Migrate and Differentiate into Mature Neurons in the Post-Stroke Adult Striatum. J. Neurosci. *26*, 6627–6636. https://doi.org/10.1523/JNEUROSCI.0149-06.2006.
22. Inta, D., Cameron, H.A., and Gass, P. (2015). New neurons in the adult striatum: from rodents to humans. Trends Neurosci. *38*, 517–523. https://doi.org/10.1016/j.tins.2015.07.005.
23. Buchanan, J., Elabbady, L., Collman, F., Jorstad, N.L., Bakken, T.E., Ott, C., Glatzer, J., Bleckert, A.A., Bodor, A.L., Brittain, D., et al. (2022). Oligodendrocyte precursor cells ingest axons in the mouse neocortex. Proc. Natl. Acad. Sci. USA *119*, e2202580119. https://doi.org/10.1073/pnas.2202580119.
24. Alvarez-Buylla, A. (1990). Mechanism of neurogenesis in adult avian brain. Experientia *46*, 948–955. https://doi.org/10.1007/BF01939388.
25. Rother, A., Januszewski, M., Jain, V., Fee, M.S., and Kornfeld, J. (2025). The songbird basal ganglia connectome. Preprint at bioRxiv. https://doi.org/10.1101/2025.10.25.684569.
26. Schubert, P.J., Dorkenwald, S., Januszewski, M., Klimesch, J., Svara, F., Mancu, A., Ahmad, H., Fee, M.S., Jain, V., and Kornfeld, J. (2022). SyConn2: dense synaptic connectivity inference for volume electron microscopy. Nat. Methods *19*, 1367–1370. https://doi.org/10.1038/s41592-022-01624-x.
27. Barami, K., Iversen, K., Furneaux, H., and Goldman, S.A. (1995). Hu protein as an early marker of neuronal phenotypic differentiation by subependymal zone cells of the adult songbird forebrain. J. Neurobiol. *28*, 82–101. https://doi.org/10.1002/neu.480280108.
28. Alvarez-Buylla, A., García-Verdugo, J.M., Mateo, A.S., and Merchant-Larios, H. (1998). Primary Neural Precursors and Intermitotic Nuclear Migration in the Ventricular Zone of Adult Canaries. J. Neurosci. *18*, 1020–1037. https://doi.org/10.1523/JNEUROSCI.18-03-01020.1998.
29. Bovetti, S., Hsieh, Y.C., Bovolin, P., Perroteau, I., Kazunori, T., and Puche, A.C. (2007). Blood Vessels Form a Scaffold for Neuroblast Migration in the Adult Olfactory Bulb. J. Neurosci. *27*, 5976–5980. https://doi.org/10.1523/JNEUROSCI.0678-07.2007.
30. Paredes, M.F., James, D., Gil-Perotin, S., Kim, H., Cotter, J.A., Ng, C., Sandoval, K., Rowitch, D.H., Xu, D., McQuillen, P.S., et al. (2016). Extensive migration of young neurons into the infant human frontal lobe. Science *354*, aaf7073. https://doi.org/10.1126/science.aaf7073.
31. O'Rourke, N.A., Sullivan, D.P., Kaznowski, C.E., Jacobs, A.A., and McConnell, S.K. (1995). Tangential migration of neurons in the developing cerebral cortex. Dev. Camb. Engl. *121*, 2165–2176. https://doi.org/10.1242/dev.121.7.2165.
32. Alderman, P.J., Saxon, D., Torrijos-Saiz, L.I., Sharief, M., Page, C.E., Baroudi, J.K., Biagiotti, S.W., Butyrkin, V.A., Melamed, A., Kuo, C.T., et al. (2024). Delayed maturation and migration of excitatory neurons in the juvenile mouse paralaminar amygdala. Neuron *112*, 574–592. https://doi.org/10.1016/j.neuron.2023.11.010.
33. Shvedov, N.R., Analoui, S., Dafalias, T., Bedell, B.L., Gardner, T.J., and Scott, B.B. (2024). In vivo imaging in transgenic songbirds reveals superdiffusive neuron migration in the adult brain. Cell Rep. *43*, 113759. https://doi.org/10.1016/j.celrep.2024.113759.
34. Rao, M.S., and Shetty, A.K. (2004). Efficacy of doublecortin as a marker to analyse the absolute number and dendritic growth of newly generated neurons in the adult dentate gyrus. Eur. J. Neurosci. *19*, 234–246. https://doi.org/10.1111/j.0953-816X.2003.03123.x.
35. Balthazart, J., Boseret, G., Konkle, A.T.M., Hurley, L.L., and Ball, G.F. (2008). Doublecortin as a marker of adult neuroplasticity in the canary song control nucleus HVC. Eur. J. Neurosci. *27*, 801–817. https://doi.org/10.1111/j.1460-9568.2008.06059.x.
36. Marusich, M.F., Furneaux, H.M., Henion, P.D., and Weston, J.A. (1994). Hu neuronal proteins are expressed in proliferating neurogenic cells. J. Neurobiol. *25*, 143–155. https://doi.org/10.1002/neu.480250206.
37. Scott, B.B., Gardner, T., Ji, N., Fee, M.S., and Lois, C. (2012). Wandering Neuronal Migration in the Postnatal Vertebrate Forebrain. J. Neurosci. *32*, 1436–1446. https://doi.org/10.1523/JNEUROSCI.2145-11.2012.
38. Bellion, A., Baudoin, J.P., Alvarez, C., Bornens, M., and Métin, C. (2005). Nucleokinesis in Tangentially Migrating Neurons Comprises Two Alternating Phases: Forward Migration of the Golgi/Centrosome Associated with Centrosome Splitting and Myosin Contraction at the Rear. J. Neurosci. *25*, 5691–5699. https://doi.org/10.1523/JNEUROSCI.1030-05.2005.
39. Kappeler, C., Saillour, Y., Baudoin, J.P., Tuy, F.P.D., Alvarez, C., Houbron, C., Gaspar, P., Hamard, G., Chelly, J., Métin, C., et al. (2006). Branching and nucleokinesis defects in migrating interneurons derived from doublecortin knockout mice. Hum. Mol. Genet. *15*, 1387–1400. https://doi.org/10.1093/hmg/ddl062.
40. Matsumoto, M., Sawada, M., García-González, D., Herranz-Pérez, V., Ogino, T., Bang Nguyen, H., Quynh Thai, T., Narita, K., Kumamoto, N., Ugawa, S., et al. (2019). Dynamic Changes in Ultrastructure of the Primary Cilium in Migrating Neuroblasts in the Postnatal Brain. J. Neurosci. *39*, 9967–9988. https://doi.org/10.1523/JNEUROSCI.1503-19.2019.

41. Schaar, B.T., and McConnell, S.K. (2005). Cytoskeletal coordination during neuronal migration. Proc. Natl. Acad. Sci. USA *102*, 13652–13657. https://doi.org/10.1073/pnas.0506008102.

42. Yanagida, M., Miyoshi, R., Toyokuni, R., Zhu, Y., and Murakami, F. (2012). Dynamics of the leading process, nucleus, and Golgi apparatus of migrating cortical interneurons in living mouse embryos. Proc. Natl. Acad. Sci. USA *109*, 16737–16742. https://doi.org/10.1073/pnas.1209166109.

43. Mascanzoni, F., Iannitti, R., and Colanzi, A. (2022). Functional Coordination among the Golgi Complex, the Centrosome and the Microtubule Cytoskeleton during the Cell Cycle. Cells *11*, 354. https://doi.org/10.3390/cells11030354.

44. Nakazawa, N., and Kengaku, M. (2020). Mechanical Regulation of Nuclear Translocation in Migratory Neurons. Front. Cell Dev. Biol. *8*, 150. https://doi.org/10.3389/fcell.2020.00150.

45. Nakajima, C., Sawada, M., Umeda, E., Takagi, Y., Nakashima, N., Kuboyama, K., Kaneko, N., Yamamoto, S., Nakamura, H., Shimada, N., et al. (2024). Identification of the growth cone as a probe and driver of neuronal migration in the injured brain. Nat. Commun. *15*, 1877. https://doi.org/10.1038/s41467-024-45825-8.

46. Chang, D.T.W., and Reynolds, I.J. (2006). Differences in mitochondrial movement and morphology in young and mature primary cortical neurons in culture. Neuroscience *141*, 727–736. https://doi.org/10.1016/j.neuroscience.2006.01.034.

47. Lin-Hendel, E.G., McManus, M.J., Wallace, D.C., Anderson, S.A., and Golden, J.A. (2016). Differential Mitochondrial Requirements for Radially and Non-radially Migrating Cortical Neurons: Implications for Mitochondrial Disorders. Cell Rep. *15*, 229–237. https://doi.org/10.1016/j.celrep.2016.03.024.

48. Varum, S., Rodrigues, A.S., Moura, M.B., Momcilovic, O., Easley, C.A., Ramalho-Santos, J., Van Houten, B., and Schatten, G. (2011). Energy Metabolism in Human Pluripotent Stem Cells and Their Differentiated Counterparts. PLoS One *6*, e20914. https://doi.org/10.1371/journal.pone.0020914.

49. St John, J.C., Ramalho-Santos, J., Gray, H.L., Petrosko, P., Rawe, V.Y., Navara, C.S., Simerly, C.R., and Schatten, G.P. (2005). The expression of mitochondrial DNA transcription factors during early cardiomyocyte in vitro differentiation from human embryonic stem cells. Cloning Stem Cells *7*, 141–153. https://doi.org/10.1089/clo.2005.7.141.

50. Almeida, A.S., and Vieira, H.L.A. (2017). Role of Cell Metabolism and Mitochondrial Function During Adult Neurogenesis. Neurochem. Res. *42*, 1787–1794. https://doi.org/10.1007/s11064-016-2150-3.

51. Houghton, F.D. (2006). Energy metabolism of the inner cell mass and trophectoderm of the mouse blastocyst. Differentiation *74*, 11–18. https://doi.org/10.1111/j.1432-0436.2006.00052.x.

52. Bertholet, A.M., Millet, A.M.E., Guillermin, O., Daloyau, M., Davezac, N., Miquel, M.C., and Belenguer, P. (2013). OPA1 loss of function affects in vitro neuronal maturation. Brain *136*, 1518–1533. https://doi.org/10.1093/brain/awt060.

53. Alvarez-Buylla, A., Theelen, M., and Nottebohm, F. (1988). Mapping of radial glia and of a new cell type in adult canary brain. J. Neurosci. *8*, 2707–2712. https://doi.org/10.1523/JNEUROSCI.08-08-02707.1988.

54. Burd, G.D., and Nottebohm, F. (1985). Ultrastructural characterization of synaptic terminals formed on newly generated neurons in a song control nucleus of the adult canary forebrain. J. Comp. Neurol. *240*, 143–152. https://doi.org/10.1002/cne.902400204.

55. Fortune, E.S., and Margoliash, D. (1992). Cytoarchitectonic organization and morphology of cells of the field L complex in male zebra finches (taenopygia guttata). J. Comp. Neurol. *325*, 388–404. https://doi.org/10.1002/cne.903250306.

56. Gahr, M., and Garcia-Segura, L.M. (1996). Testosterone-dependent increase of gap-junctions in HVC neurons of adult female canaries. Brain Res. *712*, 69–73. https://doi.org/10.1016/0006-8993(95)01448-9.

57. Alcami, P., Totagera, S., Sohnius-Wilhelmi, N., Leitner, S., Grothe, B., Frankl-Vilches, C., and Gahr, M. (2021). Extensive GJD2 Expression in the Song Motor Pathway Reveals the Extent of Electrical Synapses in the Songbird Brain. Biology *10*, 1099. https://doi.org/10.3390/biology10111099.

58. Gour, A., Boergens, K.M., Heike, N., Hua, Y., Laserstein, P., Song, K., and Helmstaedter, M. (2021). Postnatal connectomic development of inhibition in mouse barrel cortex. Science *371*, eabb4534. https://doi.org/10.1126/science.abb4534.

59. Cordero Cervantes, D., Khare, H., Wilson, A.M., Mendoza, N.D., Coulon-Mahdi, O., Lichtman, J.W., and Zurzolo, C. (2023). 3D reconstruction of the cerebellar germinal layer reveals tunneling connections between developing granule cells. Sci. Adv. *9*, eadf3471. https://doi.org/10.1126/sciadv.adf3471.

60. Liang, Y., Li, K., Riecken, K., Maslyukov, A., Gomez-Nicola, D., Kovalchuk, Y., Fehse, B., and Garaschuk, O. (2016). Long-term in vivo single-cell tracking reveals the switch of migration patterns in adult-born juxtaglomerular cells of the mouse olfactory bulb. Cell Res. *26*, 805–821. https://doi.org/10.1038/cr.2016.55.

61. Sorrells, S.F., Paredes, M.F., Velmeshev, D., Herranz-Pérez, V., Sandoval, K., Mayer, S., Chang, E.F., Insausti, R., Kriegstein, A.R., Rubenstein, J.L., et al. (2019). Immature excitatory neurons develop during adolescence in the human amygdala. Nat. Commun. *10*, 2748. https://doi.org/10.1038/s41467-019-10765-1.

62. Ang, E.S.B.C., Haydar, T.F., Gluncic, V., and Rakic, P. (2003). Four-dimensional migratory coordinates of GABAergic interneurons in the developing mouse cortex. J. Neurosci. *23*, 5805–5815. https://doi.org/10.1523/JNEUROSCI.23-13-05805.2003.

63. Inada, H., Watanabe, M., Uchida, T., Ishibashi, H., Wake, H., Nemoto, T., Yanagawa, Y., Fukuda, A., and Nabekura, J. (2011). GABA Regulates the Multidirectional Tangential Migration of GABAergic Interneurons in Living Neonatal Mice. PLoS One *6*, e27048. https://doi.org/10.1371/journal.pone.0027048.

64. Tinterri, A., Menardy, F., Diana, M.A., Lokmane, L., Keita, M., Coulpier, F., Lemoine, S., Mailhes, C., Mathieu, B., Merchan-Sala, P., et al. (2018). Active intermixing of indirect and direct neurons builds the striatal mosaic. Nat. Commun. *9*, 4725. https://doi.org/10.1038/s41467-018-07171-4.

65. Lois, C., García-Verdugo, J.-M., and Alvarez-Buylla, A. (1996). Chain Migration of Neuronal Precursors. Science *271*, 978–981. https://doi.org/10.1126/science.271.5251.978.

66. Paul, C.D., Mistriotis, P., and Konstantopoulos, K. (2017). Cancer cell motility: lessons from migration in confined spaces. Nat. Rev. Cancer *17*, 131–140. https://doi.org/10.1038/nrc.2016.123.

67. Tyler, W.J. (2012). The mechanobiology of brain function. Nat. Rev. Neurosci. *13*, 867–878. https://doi.org/10.1038/nrn3383.

68. Kasuba, K.C., Buccino, A.P., Bartram, J., Gaub, B.M., Fauser, F.J., Ronchi, S., Kumar, S.S., Geissler, S., Nava, M.M., Hierlemann, A., et al. (2024). Mechanical stimulation and electrophysiological monitoring at subcellular resolution reveals differential mechanosensation of neurons within networks. Nat. Nanotechnol. *19*, 825–833. https://doi.org/10.1038/s41565-024-01609-1.

69. Cepkenovic, B., Friedland, F., Noetzel, E., Maybeck, V., and Offenhäusser, A. (2023). Single-neuron mechanical perturbation evokes calcium plateaus that excite and modulate the network. Sci. Rep. *13*, 20669. https://doi.org/10.1038/s41598-023-47090-z.

70. Wang, B., Torok, Z., Duffy, A., Bell, D.G., Wongso, S., Velho, T.A.F., Fairhall, A.L., and Lois, C. (2024). Unsupervised restoration of a complex learned behavior after large-scale neuronal perturbation. Nat. Neurosci. *27*, 1176–1186. https://doi.org/10.1038/s41593-024-01630-6.

71. Duffy, A., Abe, E., Perkel, D.J., and Fairhall, A.L. (2019). Variation in sequence dynamics improves maintenance of stereotyped behavior in an example from bird song. Proc. Natl. Acad. Sci. USA *116*, 9592–9597. https://doi.org/10.1073/pnas.1815910116.

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72. Dalm, S., van Gerven, M., and Ahmad, N. (2025). Effective Learning with Node Perturbation in Multi-layer Neural Networks. Preprint at arXiv. https://doi.org/10.48550/arXiv.2310.00965.

73. D'Oro, P., Schwarzer, M., Nikishin, E., Bacon, P.L., Bellemare, M.G., and Courville, A. (2022). Sample-Efficient Reinforcement Learning by Breaking the Replay Ratio Barrier. In Deep reinforcement learning workshop, NeurIPS 2022 (OpenReview). https://openreview.net/forum?id=4GBGwVIEYJ.

74. Cragg, B. (1980). Preservation of extracellular space during fixation of the brain for electron microscopy. Tissue Cell *12*, 63–72. https://doi.org/10.1016/0040-8166(80)90052-X.

75. Vellema, M., van der Linden, A., and Gahr, M. (2010). Area-specific migration and recruitment of new neurons in the adult songbird brain. J. Comp. Neurol. *518*, 1442–1459. https://doi.org/10.1002/cne.22281.

76. Nottebohm, F., Stokes, T.M., and Leonard, C.M. (1976). Central control of song in the canary, Serinus canarius. J. Comp. Neurol. *165*, 457–486. https://doi.org/10.1002/cne.901650405.

77. Stokes, T.M., Leonard, C.M., and Nottebohm, F. (1974). The telencephalon, diencephalon, and mesencephalon of the canary, Serinus canaria, in stereotaxic coordinates. J. Comp. Neurol. *156*, 337–374. https://doi.org/10.1002/cne.901560305.

78. Holzenberger, M., Jarvis, E.D., Chong, C., Grossman, M., Nottebohm, F., and Scharff, C. (1997). Selective Expression of Insulin-Like Growth Factor II in the Songbird Brain. J. Neurosci. *17*, 6974–6987. https://doi.org/10.1523/JNEUROSCI.17-18-06974.1997.

79. Scharff, C., Kirn, J.R., Grossman, M., Macklis, J.D., and Nottebohm, F. (2000). Targeted Neuronal Death Affects Neuronal Replacement and Vocal Behavior in Adult Songbirds. Neuron *25*, 481–492. https://doi.org/10.1016/S0896-6273(00)80910-1.

80. Alvarez-Buylla, A., and Kirn, J.R. (1997). Birth, migration, incorporation, and death of vocal control neurons in adult songbirds. J. Neurobiol. *33*, 585–601. https://doi.org/10.1002/(SICI)1097-4695(19971105)33:5<585::AID-NEU7>3.0.CO;2-0.

81. Ikeda, M.Z., Jeon, S.D., Cowell, R.A., and Remage-Healey, L. (2015). Norepinephrine Modulates Coding of Complex Vocalizations in the Songbird Auditory Cortex Independent of Local Neuroestrogen Synthesis. J. Neurosci. *35*, 9356–9368. https://doi.org/10.1523/JNEUROSCI.4445-14.2015.

82. Agate, R.J., Scott, B.B., Haripal, B., Lois, C., and Nottebohm, F. (2009). Transgenic songbirds offer an opportunity to develop a genetic model for vocal learning. Proc. Natl. Acad. Sci. USA *106*, 17963–17967. https://doi.org/10.1073/pnas.0909139106.

83. Kim, Y.-H., Peregrine, J., and Arnold, A.P. (2006). The distribution of expression of doublecortin (DCX) mRNA and protein in the zebra finch brain. Brain Res. *1106*, 189–196. https://doi.org/10.1016/j.brainres.2006.05.080.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Data		
Zebra finch connectome	https://syconn.esc.mpcdf.mpg.de/	j0251_72_seg_20210127_agglo2_syn_20220811_celltypes_20230822
Antibodies		
mouse monoclonal anti-DCX	Santa Cruz Biotechnology	sc-271390 (E−6), lot: C3121; RRID: AB_10610966
mouse monoclonal anti-HuC/HuD	ThermoFisher Scientific	A-21271; RRID: AB_221448
Alexa Fluor 568 conjugated goat anti-mouse	Invitrogen	RRID: 2534072
Vectashield Mounting Medium with DAPI	Vector Laboratories	H-1200-10; RRID: AB_2336790
Experimental models: Organisms/strains		
Zebra Finch UBC-GFP	Shvedov et al.[33]	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Experimental model details

Adult male zebra finches (>90 days) from the UBC-GFP strain[33,82] were bred in-house at Boston University. Birds were housed under standard conditions (12h/12h light/dark cycle). Animal use procedures were approved by the Boston University Institutional Animal Care and Use Committee (IACUC; Protocol #201800577) and carried out in accordance with National Institutes of Health standards.

METHOD DETAILS

Immunohistochemistry

Finches were euthanized with Euthasol (Virbac) and perfused with up to 20 mL of 1X phosphate-buffered saline (PBS), followed by up to 50 mL of 10% neutral buffered formalin. The brain was extracted and stored in formalin overnight. After fixation, the brain was washed with PBS and sliced into 40 or 50 μm sections with a vibratome (Leica; VT1000S).

The primary antibodies used were a mouse monoclonal anti-DCX[83] (1:300 dilution) (Santa Cruz Biotechnology, sc-271390 (E−6), lot: C3121). We also stained with a mouse monoclonal anti-HuC/HuD[27] that is a marker for cells in neuronal lineage in 1:250 dilution (ThermoFisher Scientific A-21271). The secondary antibodies used were Alexa Fluor 568 conjugated goat anti-mouse (Invitrogen) in a 1:500 dilution.

Antigen retrieval was performed prior to staining by incubating slices in a 50 mM solution of Tris buffer (Fisher Scientific) at a pH of 8.4 for 30 min. Then, tissue sections were placed in a blocking solution composed of 2% nonfat dry milk, 0.02% Triton X-100 (Sigma), and 1X PBS. Sections were incubated at 4°C overnight in primary antibody diluted in blocking solution, washed 3 times with PBS, then incubated with secondary antibody diluted in blocking solution for 1 h at room temperature or overnight at 4°C. Sections were mounted on microscope slides (Fisher Scientific) using Vectashield Antifade Mounting Medium with DAPI (Vector Laboratories, Inc.). Histological fluorescent images from striatal nucleus Area X were acquired using a confocal microscope (Zeiss LSM800, Germany). Image processing was performed with FIJI/ImageJ.

QUANTIFICATION AND STATISTICAL ANALYSIS

Machine learning based identification of migratory neurons

Ground truth cells (n=14) were manually selected from the zebra finch Area X connectome[25] using morphological characteristics shared with fluorescent imaging data identifying migratory neuron morphology (Figure S1). These cells were used to train a supervised learning algorithm, based on previous methods.[26] The IDs for these cells were: 751594441, 971410555, 2157811112, 38332384, 1773210639, 328158994, 1644151292, 561503453, 1104854425, 28511657, 2656656549, 669941616, 763200873, 245868794.

This approach returned 135 objects which were manually inspected. 40/135 exhibited morphological and ultrastructural features consistent with migratory neurons. This data indicates a false positive rate of 79.4%.

To evaluate the false negative rate, i.e. migratory neurons that were missed by the algorithm, we visually inspected a set of >2,000 low-confidence (< 0.5 certainty) medium spiny neurons and microglia, possessing a segment volume between 460

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and 1500 μm, and <200 identified synapses. Using this method, we identified 28 cells with morphology and ultrastructure features consistent with migratory neurons. All 28 cells found through a manual algorithm were found within these 40, indicating an undetectable false negative rate. If cells were drastically clipped, or had no distinct leading process (5/40), they were excluded from further analysis.

Cell orientation analysis

To determine the orientation vectors of each cell, three-dimensional X, Y, Z coordinates were extracted in the Syconn Neuroglancer interface. The coordinates corresponded to the start of the soma (a_{XYZ}), the first bifurcation point of the leading process (b_{XYZ}), and the distal tip of the leading process (c_{XYZ}; Figures 2A and 2B).

The selection of the first branch point was chosen based on the known properties of neuron migration in the adult avian brain. *In vivo* imaging studies have revealed a process of saltatory migration in which migration proceeds in a step wise fashion.[37] In the first step the leading process extends and then the nucleus of the cell invades the leading process. While further branches may serve as the leading process for nuclear invasions, these processes are highly dynamic and it would be hard to guess which direction the cell would head in. Thus selection of the first branch gives the best indication of where the nucleus will move on the next time point.

All further analyses were performed in Matlab using a custom script.

The orientation of the cell was determined by the direction of the vector between a_{XYZ} and b_{XYZ} and the magnitude of the vector (only used for plotting in Figures 2C and 2D) was determined by the approximate length of the cell from a_{XYZ} to c_{XYZ}.

After the orientation vector was extracted (b_{XYZ} - a_{XYZ}), it was normalized by dividing the vector by its Euclidean norm (using the *norm* Matlab function) to make the value of each dimension (in X, Y, and Z) its relative contribution to the overall orientation (maximum value of 1 in any dimension). If all cells were pointed in the same direction, the magnitude of their averaged orientation vector would be 1. This was to account for differences in physical length between the soma and bifurcation point of all cells which should not contribute to the overall orientation estimates.

We note that the correspondence between sample coordinates framework (XYZ) and tissue coordinates (dorsal, lateral anterior) were not recorded in the original data set.

Mitochondrial density estimates

Mitochondria were segmented and annotated previously.[25] For cell IDs that were filtered and verified above, the summed volume of all mitochondria voxels was divided by each segmented cell's volume to yield the total mitochondrial volume density ($\mu m^3/\mu m^3$). A one-way ANOVA was performed to test for significant differences in mitochondrial density across groups.

Synapse density estimates

Synapses were segmented and annotated previously using automated methods.[25,26] Only synapses with a classifier certainty greater than or equal to 0.90 were used for analysis in Matlab. For local density estimates around migratory neurons (MIGR), all synapses within a 10 μm search radius from the MIGR soma center coordinate (n = 35 MIGR somas) were counted and divided by the total search volume (20 μm x 20 μm x 20 μm) to yield the local synapse density in μm^{-3} for each cell. For local synapse density around random points, the local synapse density was computed around every micron voxel throughout the data (n = 29484).

To estimate the density of synapses in and around the path of the migrating soma, we computed the radial distribution of synapses around the soma (a_{XYZ}) and the leading process (b_{XYZ}) for each MIGR cell (n = 27, excluding 8 MIGRs that were surrounded by cell bodies of mature neurons) in Matlab. For each cell, a rotation matrix was computed using Rodrigues' rotation formula to align the leading process orientation vector ($b_{XYZ} - a_{XYZ}$) with the positive Z-axis of a cell-centered reference frame. All synapse coordinates ($\geq$ 0.90 classifier certainty) around the MIGR soma (approximately 20 x 20 x 20 μm volume centered around a_{XYZ}) were transformed into this rotated frame. A planar slab spanning 3 μm above and below the soma center was then applied, retaining only synapses whose rotated Z-coordinate fell within this range to ensure that retained synapses lie in the cross-sectional plane of the tissue directly around the MIGR soma. Radial distances from the soma center were computed in the rotated XY plane, and synapses were binned into annular rings of 0.5 μm width extending to 20 μm. Synapse density per bin was calculated as the number of synapses divided by the annular volume ($\pi(r^2_{outer} - r^2_{inner}) \times 6$ μm). Synapses that had MIGR as a pre- or post-synaptic partner (n = 29 synapses; based on classifier prediction) were excluded from all slab counts to avoid possible contribution of predicted MIGR-directed synapses to these measures. Mean synapse density and standard deviation were computed across all valid cells at each radial distance bin, omitting bins that extended beyond the EM volume for cells located along the edges. Radial synapse densities were computed similarly around the leading process measured from the center coordinate of the first bifurcation of the leading process (b_{XYZ}). For comparisons with other cell types, cell types with high frequency were randomly subsampled to match MIGR sample size. Paired Wilcoxon signed-rank and independent Wilcoxon rank-sum tests were used to test for differences in synaptic densities.

Lastly, we investigated the possibility of synapses onto or adjacent to the MIGRs. The resolution of this dataset (10 x 10 x 25 nm voxels) precluded detection of gap junctions. We did observe possible pre-synaptic terminals onto MIGRs for classifier-predicted synapses, but we were unable to detect synapses onto MIGR somas that possessed all ultrastructural features of functional synapses (pre and post-synaptic densities, vesicles). Thus, it remains unclear whether nearby mature neurons made synapses with MIGRs or whether MIGRs have the ability to detect neurotransmitter release.

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Shannon Entropy estimation

Shannon entropy of actual data (soma and synapse positions) or simulated data was computed as described previously.[33] Briefly, the three-dimensional volume containing the positions was binned into different sized bins, and we computed the probability of containing a data point for each 3-dimensional bin. We then input these probabilities into the Shannon entropy formula:

$$Entropy = -\sum p(Sb) * \log(p(Sb)), \quad \text{Equation 1}$$

where p(S*b*) is the probability that a soma's or synapse's position is within any spatial bin. We repeated this process while we increased the bin sizes and plotted the entropy across bin sizes.

Cell somas and cluster analysis

Soma positions were extracted from segmented objects with several predicted compartments. Only cells that had a soma compartment, as well as dendrites and an axon that were at least 200 μm in skeleton path length were selected. All cell types other than MSN and INT were then manually verified and any segmented cells that were merged with another fragment or cell were removed from analysis. For MSN and INT, a random subset of cells was verified manually. The median of the soma vertex coordinates was used as the "soma center." Three-dimensional coordinates from each segmented soma were then analyzed in Matlab (n = 8576 neurons).

To find the nearest neighbor of all mature neurons, pairwise distance calculations were performed between each soma center coordinate. The minimum pairwise distance for each cell was considered the distance to its nearest neighbor. A similar analysis was performed for MIGR cells for which the cell soma center was determined manually in the SyConn Neuroglancer interface. Then, the minimum pairwise distance between MIGR and all other mature neurons was extracted to find the distance to the nearest neighbor. The minimum distance found between a MIGR soma center and a mature neuron soma center was 2.575 μm.

For "random" distances, we created a binary three-dimensional matrix where the position of each soma center coordinate was a "1". Then, we performed a distance transform on this binary 3D soma matrix to find distance between every element and the nearest nonzero element. Distances below 2.575 were determined to be biologically improbable and were removed.

For an analysis of cluster size and composition, a graph-based connected components approach was taken. First, a distance transform on the binary 3D matrix of soma positions was performed. The resultant matrix contained the 3D distance between each element and every other nonzero element. We next iteratively determined an appropriate distance threshold, 9 μm (approximately double the average soma radius of 4.46 μm) to find cells belonging to the same cluster. After testing several distance thresholds ranging from 8 to 11 μm, we found that 9 μm yielded no false positives in the largest output clusters. However, using this approach, we detected a nonzero false negative rate, suggesting we are underrepresenting cluster sizes with this analysis.

Then, we applied this threshold to the distance matrix so that every value greater than 9 μm became zero, and cells that were clustered together below this distance threshold were represented as groups of ones. Using Matlab's conncomp function, connected components composed of singular groups of ones were found and stored in structs as separate clusters. The cell IDs belonging to these clusters were extracted and visualized in the Syconn Neuroglancer interface to confirm accurate cluster identification with this algorithm. Cluster sizes were computed from the data and cluster compositions were extracted.